Project: Linux-Server-Administration-Automation-on-AWS-EC2

A hands-on Linux project: setting up a multi-user server on Amazon Linux 2023 (AWS EC2) with role-based access control, text processing, process management, and a cron-automated daily system health report.

Topics Covered: Linux File System | User & Permission Management | chmod & chown | Process Management | crontab | File Navigation | Text Processing | System Information Commands | Pipes & Filters

Environment: AWS EC2 (Amazon Linux 2023) — Free Tier

Project Architecture

Linux Server (EC2 or VM)
├── Users & Groups
│ ├── alice (developer)
│ ├── bob (developer)
│ └── charlie (manager)
├── Directory Structure
│ ├── /project/dev/
│ ├── /project/manager/
│ └── /project/shared/
├── Permissions
│ ├── dev group → /project/dev/ (read/write)
│ ├── manager group → /project/manager/ (read/write)
│ └── everyone → /project/shared/ (read/write)
└── Automation
└── crontab (daily system report)

Setup — Launch Your Environment

Option A — AWS EC2 (Free Tier):

- Launch a **t2. micro Amazon Linux 2023** instance
- Security Group: allow SSH port 22 from My IP only
- Connect via SSH: ssh -i "your-key.pem" ec2-user@<public-ip>

Option B — Local VirtualBox VM:

- Open VirtualBox. Start your Linux VM (CentOS or Ubuntu)
- Open the terminal inside the VM
- All commands are the same regardless of which option you choose

Phase 1 — System Information & File System Exploration

Step 1: Gather system information

Run the following commands and screenshot each output:

Commands:

```
# Basic system info
uname -a
hostname
whoami
id
uptime
date
cal

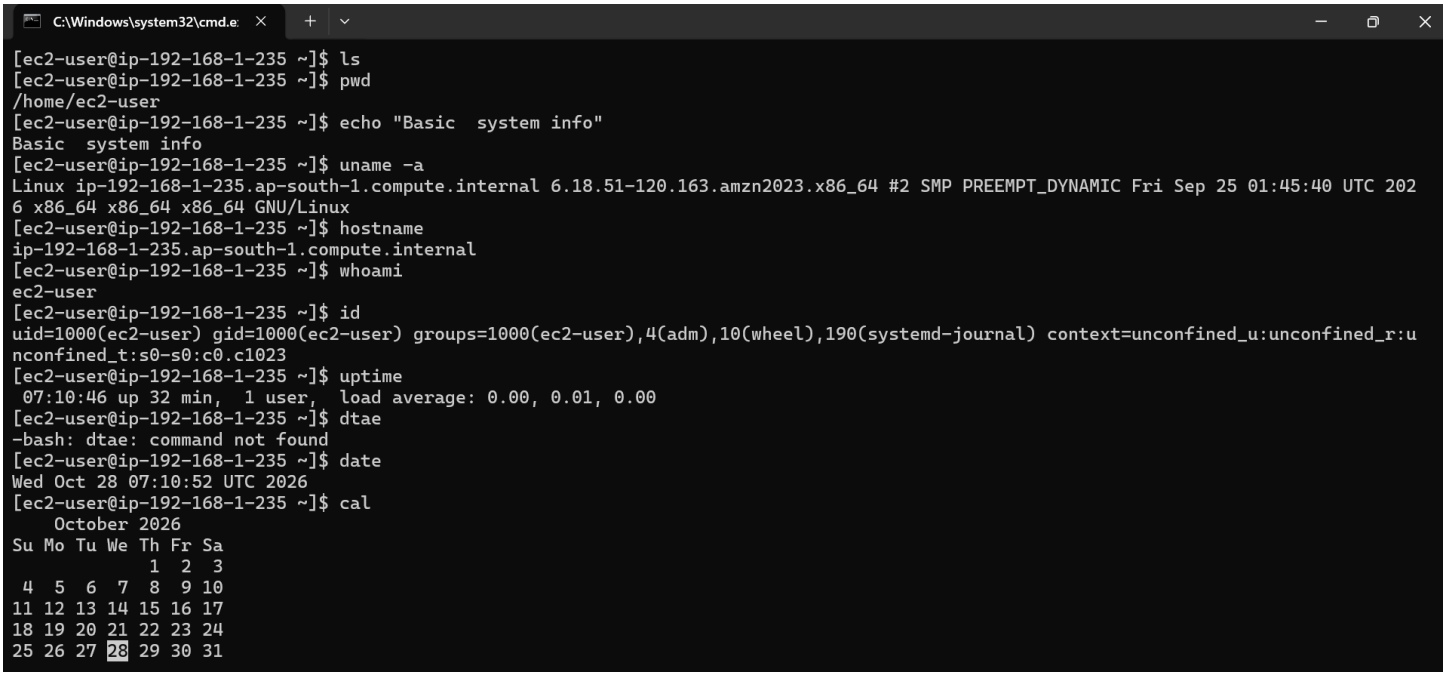
# OS information
cat /etc/os-release

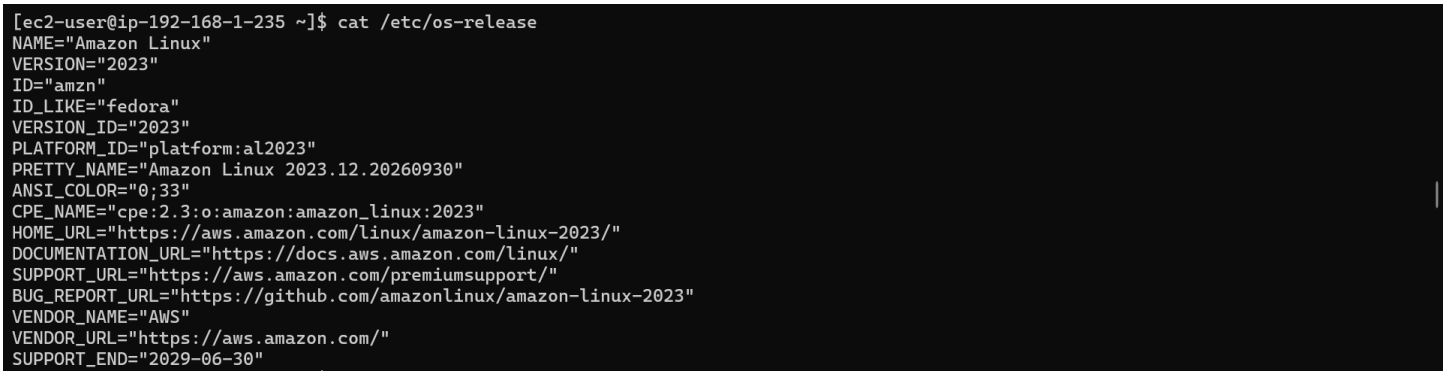
# Hardware info
nproc
free -h
df -h
lsblk
```

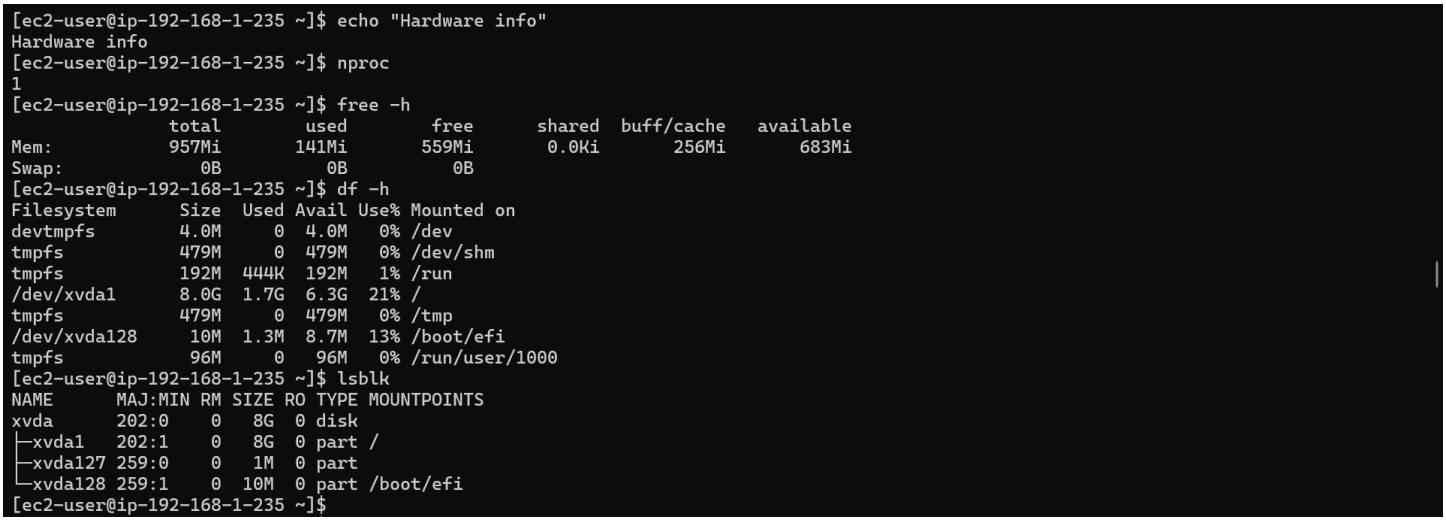
Output:

```
C:\Users\NeerajKumar\Downloads>ssh -i "practice_key.pem" ec2-user@65.0.45.231
The authenticity of host '65.0.45.231 (65.0.45.231)' can't be established.
ED25519 key fingerprint is SHA256:r0xRPNBaTAhlrFeyqLQNkZhJtQAO59oBkcIKyM46QLY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '65.0.45.231' (ED25519) to the list of known hosts.

A newer release of "Amazon Linux" is available.
  Version 2023.12.20261012:
  Version 2023.12.20261013:
  Version 2023.12.20261014:
  Version 2023.12.20261016:
  Version 2023.12.20261021:
  Version 2023.12.20261026:
Run "/usr/bin/dnf check-release-update" for full release and version update info
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~.-.   _/
         _/ _/
       _/m/'
Last login: Sun Oct 11 07:08:49 2026 from 13.233.177.3
```







Step 2: Explore the Linux file system structure

Navigate through and document the purpose of each directory:

Commands:

ls /

ls /etc

ls /var/log
ls /home
ls /tmp
ls /usr/bin

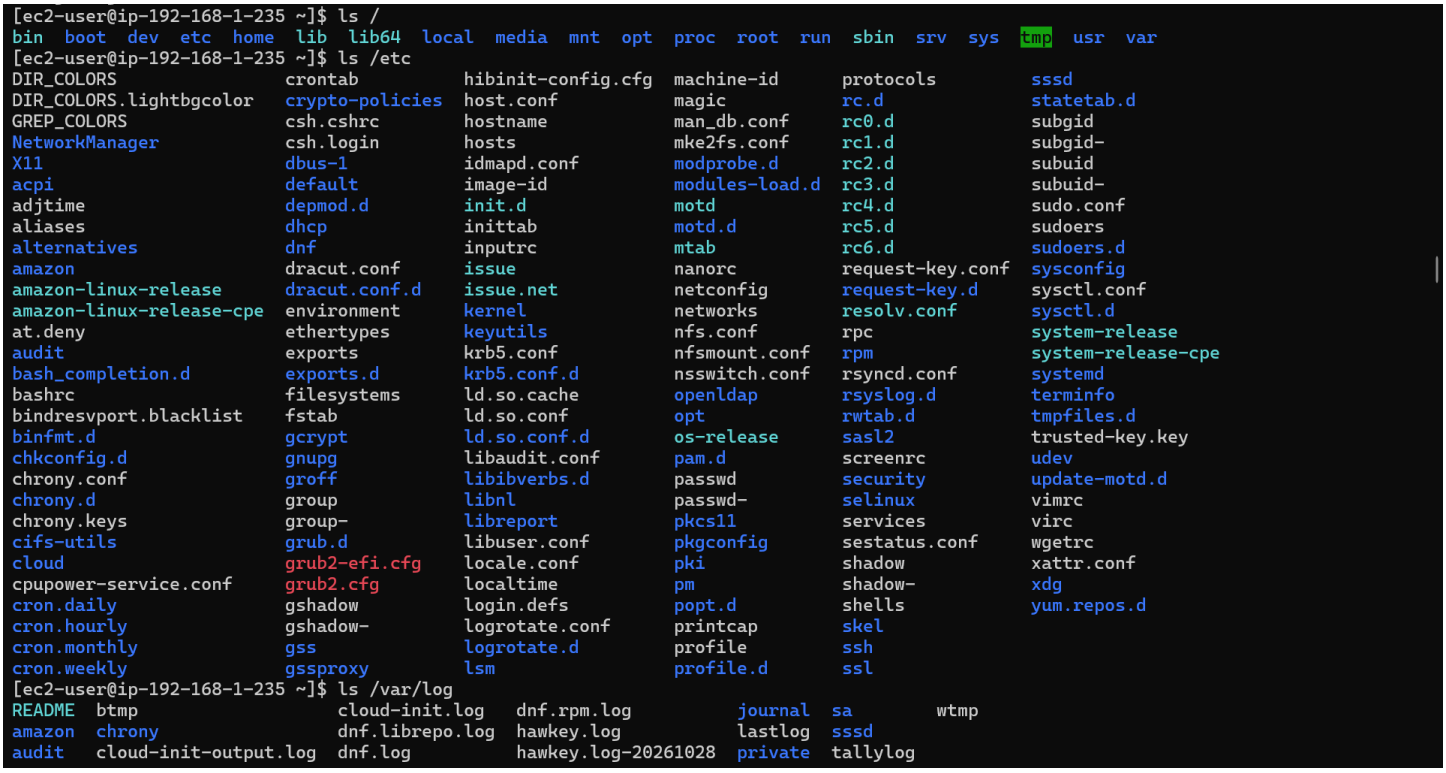
Show full path at all times
pwd

Navigate using absolute paths
cd /var/log
pwd
ls -la

Navigate using relative paths
cd ../../tmp
pwd
cd ~
pwd

Create a table in your Google Doc listing each directory and its purpose: /, /etc, /var, /home, /tmp, /usr, /bin, /sbin, /opt.

Outputs:



```
[ec2-user@ip-192-168-1-235 ~]$ ls /home
ec2-user
[ec2-user@ip-192-168-1-235 ~]$ ls /tmp
systemd-private-6e490e7e97454b26a5e03ad070bbe3ec-chronyd.service-lswdTa
systemd-private-6e490e7e97454b26a5e03ad070bbe3ec-dbus-broker.service-VV3Ky1
systemd-private-6e490e7e97454b26a5e03ad070bbe3ec-policy-routes@enX0.service-OLiflK
systemd-private-6e490e7e97454b26a5e03ad070bbe3ec-systemd-logind.service-WjvFqK
systemd-private-6e490e7e97454b26a5e03ad070bbe3ec-systemd-resolved.service-qdbQGw
[ec2-user@ip-192-168-1-235 ~]$ ls /usr/bin
'['                     gpg-error               p11-kit                 sss_ssh_knownhostsproxy
 ac                     gpg2                    package-cleanup         stap-merge
 acpi_listen            gpgme-json              passwd                  stap-report
 addr2line              gpic                    paste                   stapbpf
 alias                  gprof                   pathchk                 stapdyn
 amazon-ssm-agent       grep                    peekfd                  staprun
 apropos                groff                   perl                    stapsh
 apropos.man-db         grops                   perl5.32.1              stat
 ar                     grotty                  perldoc                 stdbuf
 arch                   groups                  pgrep                   strace
 arpaname               growpart                pic                     strace-log-merge
 arping                 grub2-amazon-setup      piconv                  strings
 as                     grub2-editenv           pidof                   strip
 at                     grub2-file              pidstat                 stty
 atq                    grub2-menulst2cfg       pidwait                 su
 atrm                   grub2-mkimage           ping                    sudo
 attr                   grub2-mkpasswd-pbkdf2   pinky                   sudoedit
 audit2allow            grub2-mkrelpath         pkg-config              sudoreplay
 audit2why              grub2-mount             pkgconf                 sum
 aulast                 grub2-script-check      pkill                   sync
 aulastlog              gsettings               pldd                    systemctl
 ausyscall              gsoelim                 pmap                    systemd-analyze
 auvirt                 gtar                    pod2man                 systemd-ask-password
 awk                    gtbl                    pod2text                systemd-cat
 aws                    gtroff                  pod2usage               systemd-cgls
 aws_completer          gunzip                  portablectl             systemd-cgtop
 b2sum                  gzexe                   post-grohtml            systemd-creds
 base32                 gzip                    powernow-k8-decode      systemd-cryptenroll
```

```
[ec2-user@ip-192-168-1-235 ~]$ pwd
/home/ec2-user
[ec2-user@ip-192-168-1-235 ~]$ cd /var/log
[ec2-user@ip-192-168-1-235 log]$ pwd
/var/log
[ec2-user@ip-192-168-1-235 log]$ ls -la
total 568
drwxr-xr-x.  9 root   root             16384 Oct 28 06:22 .
drwxr-xr-x. 19 root   root               266 Oct 10 05:23 ..
lrwxrwxrwx.  1 root   root                39 Sep 28 23:38 README -> ../../usr/share/doc/systemd/README.logs
drwxr-xr-x.  3 root   root                17 Oct 10 05:23 amazon
drwx------.  2 root   root                23 Oct 10 05:23 audit
-rw-rw----.  1 root   utmp               384 Oct 11 07:22 btmp
drwxr-x---.  2 chrony chrony             174 Oct 28 06:22 chrony
-rw-r-----.  1 root   adm              11648 Oct 28 06:38 cloud-init-output.log
-rw-r-----.  1 root   adm             487997 Oct 28 06:38 cloud-init.log
-rw-r--r--.  1 root   root              6656 Oct 28 06:38 dnf.librepo.log
-rw-r--r--.  1 root   root             13652 Oct 28 06:38 dnf.log
-rw-r--r--.  1 root   root               406 Oct 28 06:38 dnf.rpm.log
-rw-r--r--.  1 root   root               120 Oct 28 06:38 hawkey.log
-rw-r--r--.  1 root   root               180 Oct 11 07:44 hawkey.log-20261028
drwxr-sr-x+  3 root   systemd-journal     46 Oct 10 05:23 journal
-rw-rw-r--.  1 root   utmp            292292 Oct 28 07:09 lastlog
drwx------.  2 root   root                6 Sep 28 23:38 private
drwxr-xr-x.  2 root   root               42 Oct 28 06:22 sa
drwxr-x---.  2 root   root                6 Jul 14 17:05 sssd
-rw-------.  1 root   root                0 Sep 28 23:38 tallylog
-rw-rw-r--.  1 root   utmp             15360 Oct 28 07:09 wtmp
[ec2-user@ip-192-168-1-235 log]$
[ec2-user@ip-192-168-1-235 log]$ cd ../../tmp
[ec2-user@ip-192-168-1-235 tmp]$ pwd
/tmp
[ec2-user@ip-192-168-1-235 tmp]$ cd ~
[ec2-user@ip-192-168-1-235 ~]$ pwd
/home/ec2-user
[ec2-user@ip-192-168-1-235 ~]$
```

Create a table in your Google Doc listing each directory and its purpose: /, /etc, /var, /home, /tmp, /usr, /bin, /sbin, /opt.

Directory	Purpose	Examples of Contents
/	Root directory. The top-level directory that contains all other directories and files.	/etc, /home, /usr, /var

/etc	Stores system-wide configuration files for the operating system and installed services.	Network settings, user configuration, service configuration
/var	Stores variable data that changes during system operation.	Log files (/var/log), caches, mail queues, databases
/home	Contains the personal directories of regular users, including their files and settings.	/home/alex, documents, downloads, user configuration
/tmp	Stores temporary files created by applications and users. These files may be deleted automatically.	Temporary files, application working files
/usr	Contains most user-space applications, libraries, documentation, and shared resources installed on the system.	/usr/bin, /usr/lib, /usr/share
/bin	Contains essential user commands needed for basic system operation, such as commands used to manage files and run programs.	ls, cp, mv, cat
/sbin	Contains system administration utilities used for system maintenance, booting, and recovery.	fsck, reboot, ip
/opt	Used for optional or third-party application software, often installed in separate application directories.	/opt/google, /opt/myapp

Step 3: Create the project directory structure

Commands:

```
# Switch to root
sudo su -

# Create main project directory
mkdir -p /project/dev
mkdir -p /project/manager
mkdir -p /project/shared

# Verify structure
ls -la /project/
tree /project/
# Install if needed: yum install tree -y
```

Outputs:

```
[ec2-user@ip-192-168-1-235 ~]$ sudo su -
[root@ip-192-168-1-235 ~]# pwd
/root
[root@ip-192-168-1-235 ~]# mkdir -p /project/dev
[root@ip-192-168-1-235 ~]# mkdir -p /project/manager
[root@ip-192-168-1-235 ~]# mkdir -p /project/shared
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# ls -la /project/
total 0
drwxr-xr-x.  5 root root  46 Oct 28 07:17 .
dr-xr-xr-x. 19 root root 252 Oct 28 07:17 ..
drwxr-xr-x.  2 root root   6 Oct 28 07:17 dev
drwxr-xr-x.  2 root root   6 Oct 28 07:17 manager
drwxr-xr-x.  2 root root   6 Oct 28 07:17 shared
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# tree /project/
-bash: tree: command not found
[root@ip-192-168-1-235 ~]# yum install tree -y
Last metadata expiration check: 0:56:29 ago on Wed Oct 28 06:22:10 2026.
Dependencies resolved.
================================================================================
 Package              Architecture          Version                   Repository                  Size
================================================================================
Installing:
 tree                 x86_64                1.8.0-6.amzn2023.0.2      amazonlinux                 56 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 56 k
Installed size: 113 k
Downloading Packages:
tree-1.8.0-6.amzn2023.0.2.x86_64.rpm                        2.2 MB/s |  56 kB     00:00
--------------------------------------------------------------------------------
Total                                                       711 kB/s |  56 kB     00:00
Running transaction check
```

```
  Version 2023.12.20261016:
    Run the following command to upgrade to 2023.12.20261016:

      dnf upgrade --releasever=2023.12.20261016

    Release notes:
     https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.12.20261016.html

  Version 2023.12.20261021:
    Run the following command to upgrade to 2023.12.20261021:

      dnf upgrade --releasever=2023.12.20261021

    Release notes:
     https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.12.20261021.html

  Version 2023.12.20261026:
    Run the following command to upgrade to 2023.12.20261026:

      dnf upgrade --releasever=2023.12.20261026

    Release notes:
     https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.12.20261026.html

================================================================================

Installed:
  tree-1.8.0-6.amzn2023.0.2.x86_64

Complete!
[root@ip-192-168-1-235 ~]# tree /project/
/project/
├── dev
├── manager
└── shared

3 directories, 0 files
```

Phase 2 — User, Group & Permission Management

Step 4: Create users and groups

Commands:

Create groups

groupadd developers
groupadd managers

Create users
useradd -m -g developers -s /bin/bash alice
useradd -m -g developers -s /bin/bash bob
useradd -m -g managers -s /bin/bash charlie

Set passwords
echo "alice:Alice@123" | chpasswd
echo "bob:Bob@123" | chpasswd
echo "charlie:Charlie@123" | chpasswd

Verify users created
cat /etc/passwd | grep -E "alice|bob|charlie"
cat /etc/group | grep -E "developers|managers"

Check user IDs
id alice
id bob
id charlie

Outputs:

```
[root@ip-192-168-1-235 ~]# groupadd developers
[root@ip-192-168-1-235 ~]# groupadd managers
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# useradd -m -g developers -s /bin/bash alice
[root@ip-192-168-1-235 ~]# useradd -m -g developers -s /bin/bash bob
[root@ip-192-168-1-235 ~]# useradd -m -g managers -s /bin/bash charlie
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# echo "Set passwords"
Set passwords
[root@ip-192-168-1-235 ~]# echo "alice:Alice@123" | chpasswd
[root@ip-192-168-1-235 ~]# echo "bob:Bob@123" | chpa
sswd
[root@ip-192-168-1-235 ~]# echo "charlie:Charlie@123
" | chpasswd
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# echo "Verify users created"
Verify users created
[root@ip-192-168-1-235 ~]# cat /etc/passwd | grep -E
 "alice|bob|Charlie"
alice:x:1001:1001::/home/alice:/bin/bash
bob:x:1002:1001::/home/bob:/bin/bash
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# cat /etc/group | grep -E
"developers|managers"
developers:x:1001:
managers:x:1002:
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# id alice
uid=1001(alice) gid=1001(developers) groups=1001(developers)
[root@ip-192-168-1-235 ~]# id bob
uid=1002(bob) gid=1001(developers) groups=1001(developers)
[root@ip-192-168-1-235 ~]# id charlie
uid=1003(charlie) gid=1002(managers) groups=1002(managers)
[root@ip-192-168-1-235 ~]#
```

Step 5: Set directory permissions

Commands:

Set ownership of directories
chown -R root:developers /project/dev
chown -R root:managers /project/manager
chown -R root:root /project/shared

Set permissions
chmod 770 /project/dev # owner+group: full, others: none
chmod 770 /project/manager # owner+group: full, others: none
chmod 777 /project/shared # everyone: full access

Verify permissions
ls -la /project/

Display permissions in detail
stat /project/dev
stat /project/manager
stat /project/shared

Outputs:

```
[root@ip-192-168-1-235 ~]# echo " Set ownership of directory"
 Set ownership of directory
[root@ip-192-168-1-235 ~]# chown -R root:developers /project/dev
[root@ip-192-168-1-235 ~]# chown -R root:managers /p
roject/manager
[root@ip-192-168-1-235 ~]# chown -R root:root /proje
ct/shared
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# chmod 770 /project/dev
[root@ip-192-168-1-235 ~]# chmod 770 /project/manager
[root@ip-192-168-1-235 ~]# chmod 770 /project/shared
[root@ip-192-168-1-235 ~]# chmod 777 /project/shared
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# echo "verify permissions"
verify permissions
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# ls -la /project/
total 0
drwxr-xr-x.  5 root root        46 Oct 28 07:17 .
dr-xr-xr-x. 19 root root       252 Oct 28 07:17 ..
drwxrwx---.  2 root developers   6 Oct 28 07:17 dev
drwxrwx---.  2 root managers     6 Oct 28 07:17 manager
drwxrwxrwx.  2 root root         6 Oct 28 07:17 shared
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# echo "Display permissions in details"
Display permissions in details
[root@ip-192-168-1-235 ~]# stat /project/dev
  File: /project/dev
  Size: 6               Blocks: 0          IO Block: 4096   directory
Device: ca01h/51713d    Inode: 532421      Links: 2
Access: (0770/drwxrwx---)  Uid: (    0/    root)   Gid: ( 1001/developers)
Context: unconfined_u:object_r:default_t:s0
Access: 2026-10-28 07:17:21.951644512 +0000
Modify: 2026-10-28 07:17:21.951644512 +0000
Change: 2026-10-28 07:35:34.235935619 +0000
 Birth: 2026-10-28 07:17:21.951644512 +0000
[root@ip-192-168-1-235 ~]# stat /project/manager
```

```
[root@ip-192-168-1-235 ~]# stat /project/manager
  File: /project/manager
  Size: 6               Blocks: 0          IO Block: 4096   directory
Device: ca01h/51713d    Inode: 8533319     Links: 2
Access: (0770/drwxrwx---)  Uid: (    0/    root)   Gid: ( 1002/managers)
```

```
Context: unconfined_u:object_r:default_t:s0
Access: 2026-10-28 07:17:33.801582389 +0000
Modify: 2026-10-28 07:17:33.801582389 +0000
Change: 2026-10-28 07:35:41.605897129 +0000
 Birth: 2026-10-28 07:17:33.801582389 +0000
[root@ip-192-168-1-235 ~]# stat /project/shared
  File: /project/shared
  Size: 6               Blocks: 0          IO Block: 4096   directory
Device: ca01h/51713d    Inode: 532416      Links: 2
Access: (0777/drwxrwxrwx)  Uid: (    0/    root)   Gid: (    0/    root)
Context: unconfined_u:object_r:default_t:s0
Access: 2026-10-28 07:17:42.561536676 +0000
Modify: 2026-10-28 07:17:42.561536676 +0000
Change: 2026-10-28 07:35:55.365825043 +0000
 Birth: 2026-10-28 07:17:42.561536676 +0000
[root@ip-192-168-1-235 ~]#
```

Step 6: Test permissions as each user

Commands:

Switch to alice (developer)
su - alice
touch /project/dev/alice_file.txt # should succeed
touch /project/manager/alice_file.txt # should FAIL
touch /project/shared/alice_shared.txt # should succeed
exit

Switch to charlie (manager)
su - charlie
touch /project/manager/charlie_report.txt # should succeed
touch /project/dev/charlie_file.txt # should FAIL
touch /project/shared/charlie_shared.txt # should succeed
exit

Screenshot all permission denied errors and successful operations.

Outputs:

```
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# echo "Switch to alice"
Switch to alice
[root@ip-192-168-1-235 ~]# su - alice
[alice@ip-192-168-1-235 ~]$ pwd
/home/alice
[alice@ip-192-168-1-235 ~]$ touch /project/dev/alice_file.txt
[alice@ip-192-168-1-235 ~]$ touch /project/manager/a
lice_file.txt
touch: cannot touch '/project/manager/alice_file.txt': Permission denied
[alice@ip-192-168-1-235 ~]$ touch /project/shared/al
ice_shared.txt
[alice@ip-192-168-1-235 ~]$
[alice@ip-192-168-1-235 ~]$ echo " switch to charlie"
 switch to charlie
[alice@ip-192-168-1-235 ~]$ su - charlie
Password:
[charlie@ip-192-168-1-235 ~]$ pwd
/home/charlie
[charlie@ip-192-168-1-235 ~]$ touch /project/mamager/charlie_report.txt
touch: cannot touch '/project/mamager/charlie_report.txt': No such file or directory
[charlie@ip-192-168-1-235 ~]$ touch /project/manager
/charlie_report.txt
[charlie@ip-192-168-1-235 ~]$ touch /project/dev/charlie_file.txt
touch: cannot touch '/project/dev/charlie_file.txt': Permission denied
[charlie@ip-192-168-1-235 ~]$ touch /project/shared/charlie_shared.txt
[charlie@ip-192-168-1-235 ~]$ exit
logout
```

Step 7: Work with ACL (Access Control List)

Commands:

Give charlie read access to /project/dev without changing group
setfacl -m u:charlie:r-x /project/dev
getfacl /project/dev

Verify charlie can now read dev directory
su - charlie
ls /project/dev # should now work
touch /project/dev/test.txt # should still FAIL (read only)
exit

Outputs:

```
[root@ip-192-168-1-235 ~]# setfacl -m u:charlie:r-x /project/dev
[root@ip-192-168-1-235 ~]# getfacl /project/dev
getfacl: Removing leading '/' from absolute path names
# file: project/dev
# owner: root
# group: developers
user::rwx
user:charlie:r-x
group::rwx
mask::rwx
other::---

[root@ip-192-168-1-235 ~]# su - charlie
Last login: Wed Oct 28 07:50:04 UTC 2026 on pts/1
[charlie@ip-192-168-1-235 ~]$ ls /project/dev
alice_file.txt
[charlie@ip-192-168-1-235 ~]$ touch /project/dev/test.txt
touch: cannot touch '/project/dev/test.txt': Permission denied
[charlie@ip-192-168-1-235 ~]$ exit
logout
[root@ip-192-168-1-235 ~]#
```

Step 8: Configure password aging

Commands:

Set password policy for alice
chage -M 90 -m 7 -W 14 alice

View password aging info
chage -l alice
chage -l bob

View /etc/shadow for password info (first field is encrypted password)
grep -E "alice|bob|charlie" /etc/shadow

Outputs:

```
[root@ip-192-168-1-235 ~]# chage -M 90 -m 7 -W 14 alice
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# chage -l alice
Last password change                                    : Oct 28, 2026
Password expires                                        : Jan 26, 2027
Password inactive                                       : never
```

```
Account expires                                         : never
Minimum number of days between password change          : 7
Maximum number of days between password change          : 90
Number of days of warning before password expires       : 14
[root@ip-192-168-1-235 ~]# chage -l bob
Last password change                                    : Oct 28, 2026
Password expires                                        : never
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 99999
Number of days of warning before password expires       : 7
[root@ip-192-168-1-235 ~]# grep -E "alice|bob|charlie" /etc/shadow
alice:$y$j9T$gmJno/30XtRNUtxoFPHVz0$DZGfUno95f5xnNXPN/9Iig3H0zPXDD9F7Ni5Byyg5O9:20754:7:90:14:::
bob:$y$j9T$Q7/Q5F82LpgUobZgQxXyz.$44VAI..OYWtgAQD4ILpklAsWt4DioIa9qQd5Vw3epC8:20754:0:99999:7:::
charlie:$y$j9T$aBwuCDpgkoheSo0sN1kmI0$bNlJgDiOL0RcY1eJDgTmKQMgejitA4L2PCqRw1IRJB7:20754:0:99999:7:::
[root@ip-192-168-1-235 ~]#
```

Phase 3 — File Operations & Text Processing

Step 9: Create and manage project files

Commands:

Create sample files with content
cat > /project/shared/team_members.txt << EOF
alice,developer,alice@company.com,85000
bob,developer,bob@company.com,82000
charlie,manager,charlie@company.com,95000
diana,developer,diana@company.com,80000
eve,manager,eve@company.com,92000
EOF

cat > /project/shared/server_log.txt << EOF
2024-01-15 08:00:01 INFO Server started successfully
2024-01-15 08:05:22 INFO User alice logged in
2024-01-15 08:10:45 ERROR Database connection failed
2024-01-15 08:10:47 INFO Retrying database connection
2024-01-15 08:10:50 INFO Database connection restored
2024-01-15 08:30:11 WARNING High CPU usage detected: 85%
2024-01-15 09:00:00 INFO Backup started
2024-01-15 09:15:00 INFO Backup completed successfully
2024-01-15 09:30:22 ERROR Disk space below 10%
2024-01-15 10:00:01 INFO User bob logged in
EOF

Outputs:

```
[root@ip-192-168-1-235 ~]# echo "create sample file with content"
create sample file with content
[root@ip-192-168-1-235 ~]# cat> /project/shared/team_members.txt<<EOF
> alice,developer,alice@company.com,85000
bob,developer,bob@company.com,82000
charlie,manager,charlie@company.com,95000
diana,developer,diana@company.com,80000
eve,manager,eve@company.com,92000
EOF
[root@ip-192-168-1-235 ~]# cat > /project/shared/server_log.txt << EOF
> 2024-01-15 08:00:01 INFO Server started successfully
2024-01-15 08:05:22 INFO User alice logged in
2024-01-15 08:10:45 ERROR Database connection failed
2024-01-15 08:10:47 INFO Retrying database connection
2024-01-15 08:10:50 INFO Database connection restored
2024-01-15 08:30:11 WARNING High CPU usage detected: 85%
2024-01-15 09:00:00 INFO Backup started
2024-01-15 09:15:00 INFO Backup completed successfully
2024-01-15 09:30:22 ERROR Disk space below 10%
2024-01-15 10:00:01 INFO User bob logged in
EOF
```

Step 10: Text processing commands Run each command and screenshot the output:

Commands:

```
# cat - display file
cat /project/shared/team_members.txt

# grep - search for patterns
grep "ERROR" /project/shared/server_log.txt
grep "INFO" /project/shared/server_log.txt
grep -c "ERROR" /project/shared/server_log.txt   # count errors

# awk - extract specific fields
awk -F',' '{print $1, $3}' /project/shared/team_members.txt
awk -F',' '{print "Name:", $1, "| Salary:", $4}' /project/shared/team_members.txt

# sort - sort file contents
sort /project/shared/team_members.txt
sort -t',' -k4 -n /project/shared/team_members.txt   # sort by salary

# wc - word/line count
wc -l /project/shared/server_log.txt
wc -w /project/shared/team_members.txt

# uniq - find unique lines
cat /project/shared/server_log.txt | grep -o "INFO\|ERROR\|WARNING" | sort | uniq -c

# sed - stream editor
sed 's/ERROR/CRITICAL/g' /project/shared/server_log.txt
sed -n '/ERROR/p' /project/shared/server_log.txt
# pipes and filters combined
cat /project/shared/server_log.txt | grep "ERROR" | awk '{print $1, $2, $5}' | sort
```

Outputs:

```
[root@ip-192-168-1-235 ~]# echo "create sample file with content"
create sample file with content
[root@ip-192-168-1-235 ~]# cat> /project/shared/team_members.txt<<EOF
> alice,developer,alice@company.com,85000
bob,developer,bob@company.com,82000
charlie,manager,charlie@company.com,95000
diana,developer,diana@company.com,80000
eve,manager,eve@company.com,92000
EOF
[root@ip-192-168-1-235 ~]# cat > /project/shared/server_log.txt << EOF
> 2024-01-15 08:00:01 INFO Server started successfully
2024-01-15 08:05:22 INFO User alice logged in
2024-01-15 08:10:45 ERROR Database connection failed
2024-01-15 08:10:47 INFO Retrying database connection
2024-01-15 08:10:50 INFO Database connection restored
2024-01-15 08:30:11 WARNING High CPU usage detected: 85%
2024-01-15 09:00:00 INFO Backup started
2024-01-15 09:15:00 INFO Backup completed successfully
2024-01-15 09:30:22 ERROR Disk space below 10%
2024-01-15 10:00:01 INFO User bob logged in
EOF
```

```
[root@ip-192-168-1-235 ~]# cat /project/shared/team_members.txt
alice,developer,alice@company.com,85000
bob,developer,bob@company.com,82000
charlie,manager,charlie@company.com,95000
diana,developer,diana@company.com,80000
eve,manager,eve@company.com,92000
[root@ip-192-168-1-235 ~]# grep "ERROR" /project/shared/server_log.txt
2024-01-15 08:10:45 ERROR Database connection failed
2024-01-15 09:30:22 ERROR Disk space below 10%
[root@ip-192-168-1-235 ~]# grep "INFO" /project/shared/server_log.txt
2024-01-15 08:00:01 INFO Server started successfully
2024-01-15 08:05:22 INFO User alice logged in
2024-01-15 08:10:47 INFO Retrying database connection
2024-01-15 08:10:50 INFO Database connection restored
2024-01-15 09:00:00 INFO Backup started
2024-01-15 09:15:00 INFO Backup completed successfully
```

```
2024-01-15 10:00:01 INFO User bob logged in
[root@ip-192-168-1-235 ~]# grep -c "ERROR" /project/shared/server_log.txt
2
[root@ip-192-168-1-235 ~]# awk -F',' '{print $1, $3}' /project/shared/team_members.txt
alice alice@company.com
bob bob@company.com
charlie charlie@company.com
diana diana@company.com
eve eve@company.com
[root@ip-192-168-1-235 ~]# awk -F',' '{print "Name:", $1, "| Salary:", $4}' /project/shared/team_members.txt
Name: alice | Salary: 85000
Name: bob | Salary: 82000
Name: charlie | Salary: 95000
Name: diana | Salary: 80000
Name: eve | Salary: 92000
[root@ip-192-168-1-235 ~]# sort /project/shared/team_members.txt
alice,developer,alice@company.com,85000
bob,developer,bob@company.com,82000
charlie,manager,charlie@company.com,95000
diana,developer,diana@company.com,80000
eve,manager,eve@company.com,92000
[root@ip-192-168-1-235 ~]# sort -t',' -k4 -n /project/shared/team_members.txt
diana,developer,diana@company.com,80000
bob,developer,bob@company.com,82000
alice,developer,alice@company.com,85000
eve,manager,eve@company.com,92000
charlie,manager,charlie@company.com,95000
[root@ip-192-168-1-235 ~]# wc -l /project/shared/server_log.txt
10 /project/shared/server_log.txt
[root@ip-192-168-1-235 ~]# wc -w /project/shared/team_members.txt
5 /project/shared/team_members.txt
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# cat /project/shared/server_log.txt | grep -o "INFO\|ERROR\|WARNING" | sort | uniq -c
      2 ERROR
      7 INFO
      1 WARNING
[root@ip-192-168-1-235 ~]# sed 's/ERROR/CRITICAL/g' /project/shared/server_log.txt
```

```
[root@ip-192-168-1-235 ~]# sed 's/ERROR/CRITICAL/g' /project/shared/server_log.txt
2024-01-15 08:00:01 INFO Server started successfully
2024-01-15 08:05:22 INFO User alice logged in
2024-01-15 08:10:45 CRITICAL Database connection failed
2024-01-15 08:10:47 INFO Retrying database connection
2024-01-15 08:10:50 INFO Database connection restored
2024-01-15 08:30:11 WARNING High CPU usage detected: 85%
2024-01-15 09:00:00 INFO Backup started
2024-01-15 09:15:00 INFO Backup completed successfully
2024-01-15 09:30:22 CRITICAL Disk space below 10%
2024-01-15 10:00:01 INFO User bob logged in
[root@ip-192-168-1-235 ~]# sed -n '/ERROR/p' /project/shared/server_log.txt
2024-01-15 08:10:45 ERROR Database connection failed
2024-01-15 09:30:22 ERROR Disk space below 10%
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# cat /project/shared/server_log.txt | grep "ERROR" | awk '{print $1, $2, $5}' | sort
2024-01-15 08:10:45 connection
2024-01-15 09:30:22 space
[root@ip-192-168-1-235 ~]#
```

Step 11: Compression and archiving

Commands:

Copy files to compress

cp /project/shared/team_members.txt /tmp/

cp /project/shared/server_log.txt /tmp/

gzip compression
gzip /tmp/team_members.txt
ls -lh /tmp/team_members.txt.gz

gunzip
gunzip /tmp/team_members.txt.gz
ls -lh /tmp/team_members.txt

tar archiving
tar -cvf /tmp/project_backup.tar /project/shared/
ls -lh /tmp/project_backup.tar

tar with gzip compression
tar -czvf /tmp/project_backup.tar.gz /project/shared/
ls -lh /tmp/project_backup.tar.gz

Extract
tar -xzvf /tmp/project_backup.tar.gz -C /tmp/restored/
ls /tmp/restored/project/shared/

Outputs:

```
[root@ip-192-168-1-235 ~]# cp /project/shared/team_members.txt /tmp/
[root@ip-192-168-1-235 ~]# cp /project/shared/server_log.txt /tmp/
[root@ip-192-168-1-235 ~]# gzip /tmp/team_members.txt
[root@ip-192-168-1-235 ~]# ls -lh /tmp/team_members.txt.gz
-rw-r--r--. 1 root root 130 Oct 29 07:32 /tmp/team_members.txt.gz
[root@ip-192-168-1-235 ~]# gunzip /tmp/team_members.txt.gz
[root@ip-192-168-1-235 ~]# ls -lh /tmp/team_members.txt
-rw-r--r--. 1 root root 192 Oct 29 07:32 /tmp/team_members.txt
[root@ip-192-168-1-235 ~]# tar -cvf /tmp/project_backup.tar /project/shared/
tar: Removing leading `/' from member names
/project/shared/
/project/shared/alice_shared.txt
/project/shared/charlie_shared.txt
/project/shared/team_members.txt
/project/shared/server_log.txt
[root@ip-192-168-1-235 ~]# ls -lh /tmp/project_backup.tar
-rw-r--r--. 1 root root 10K Oct 29 07:32 /tmp/project_backup.tar
[root@ip-192-168-1-235 ~]# tar -czvf /tmp/project_backup.tar.gz /project/shared/
tar: Removing leading `/' from member names
/project/shared/
/project/shared/alice_shared.txt
/project/shared/charlie_shared.txt
/project/shared/team_members.txt
/project/shared/server_log.txt
[root@ip-192-168-1-235 ~]# ls -lh /tmp/project_backup.tar.gz
-rw-r--r--. 1 root root 581 Oct 29 07:32 /tmp/project_backup.tar.gz
```

```
[root@ip-192-168-1-235 ~]# tar -xzvf /tmp/project_backup.tar.gz
project/shared/
project/shared/alice_shared.txt
project/shared/charlie_shared.txt
project/shared/team_members.txt
project/shared/server_log.txt
[root@ip-192-168-1-235 ~]# ls /tmp/restored/project/shared/
ls: cannot access '/tmp/restored/project/shared/': No such file or directory
[root@ip-192-168-1-235 ~]#
```

Phase 4 — Process Management & Automation

Step 12: Process management

Commands:

View running processes
ps aux
ps aux | grep httpd
ps -ef | head -20
Top - interactive process viewer
top -b -n 1 | head -20 # non-interactive single snapshot

Install and run a background process for testing
yum install httpd -y # EC2: yum | Ubuntu: apt install apache2 -y
systemctl start httpd
systemctl status httpd
systemctl enable httpd

Find the process ID
ps aux | grep httpd
pgrep httpd

Kill and restart
systemctl stop httpd
systemctl status httpd
systemctl start httpd

Outputs:

```
[root@ip-192-168-1-235 ~]# ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  1.8 108228 17716 ?        Ss   07:16   0:00 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
root           2  0.0  0.0      0     0 ?        S    07:16   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    07:16   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-sync_wq]
root           6  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-kvfree_rcu_reclaim]
root           7  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-slub_flushwq]
root           8  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-netns]
root          10  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/0:0H-events_highpri]
root          11  0.0  0.0      0     0 ?        I    07:16   0:00 [kworker/0:1-cgroup_offline]
root          13  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-mm_percpu_wq]
root          14  0.0  0.0      0     0 ?        S    07:16   0:00 [ksoftirqd/0]
root          15  0.0  0.0      0     0 ?        I    07:16   0:00 [rcu_preempt]
root          16  0.0  0.0      0     0 ?        S    07:16   0:00 [rcu_exp_par_gp_kthread_worker/0]
root          17  0.0  0.0      0     0 ?        S    07:16   0:00 [rcu_exp_gp_kthread_worker]
root          18  0.0  0.0      0     0 ?        S    07:16   0:00 [migration/0]
root          19  0.0  0.0      0     0 ?        S    07:16   0:00 [cpuhp/0]
root          21  0.0  0.0      0     0 ?        S    07:16   0:00 [kdevtmpfs]
root          22  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-inet_frag_wq]
root          23  0.0  0.0      0     0 ?        I    07:16   0:00 [rcu_tasks_kthread]
root          24  0.0  0.0      0     0 ?        I    07:16   0:00 [rcu_tasks_rude_kthread]
root          25  0.0  0.0      0     0 ?        I    07:16   0:00 [rcu_tasks_trace_kthread]
root          26  0.0  0.0      0     0 ?        S    07:16   0:00 [kauditd]
root          28  0.0  0.0      0     0 ?        S    07:16   0:00 [khungtaskd]
root          29  0.0  0.0      0     0 ?        S    07:16   0:00 [oom_reaper]
root          30  0.0  0.0      0     0 ?        I    07:16   0:00 [kworker/u60:1-events_unbound]
root          31  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-writeback]
root          33  0.0  0.0      0     0 ?        S    07:16   0:00 [kcompactd0]
root          34  0.0  0.0      0     0 ?        SN   07:16   0:00 [khugepaged]
root          35  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-kblockd]
root          36  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-blkcg_punt_bio]
root          37  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-kintegrityd]
root          38  0.0  0.0      0     0 ?        S    07:16   0:00 [irq/9-acpi]
root          39  0.0  0.0      0     0 ?        S    07:16   0:00 [xen-balloon]
root          40  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-tpm_dev_wq]
```

```
ec2-user    3532  0.0  0.5 224096  5008 pts/0    Ss   07:39   0:00 -bash
root        3536  0.1  0.9  19616  9692 ?        Ss   07:39   0:00 /usr/lib/systemd/systemd-hostnamed
root        3557  0.0  0.8 235464  8448 pts/0    R+   07:40   0:00 sudo su -
root        3560  0.0  0.2 235464  2644 pts/1    Ss   07:40   0:00 sudo su -
root        3561  0.0  0.4 225340  4684 pts/1    S    07:40   0:00 su -
root        3562  0.0  0.5 224092  5012 pts/1    S    07:40   0:00 -bash
root        3584  0.0  0.2 223608  2936 pts/1    R+   07:40   0:00 ps aux
[root@ip-192-168-1-235 ~]# ps aux | grep httpd
root        3588  0.0  0.2 222344  2180 pts/1    S+   07:40   0:00 grep --color=auto httpd
[root@ip-192-168-1-235 ~]# ps -ef | head -20
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 07:16 ?        00:00:00 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
root           2       0  0 07:16 ?        00:00:00 [kthreadd]
root           3       2  0 07:16 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 07:16 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 07:16 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 07:16 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 07:16 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 07:16 ?        00:00:00 [kworker/R-netns]
root          10       2  0 07:16 ?        00:00:00 [kworker/0:0H-events_highpri]
root          11       2  0 07:16 ?        00:00:00 [kworker/0:1-cgroup_offline]
root          13       2  0 07:16 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 07:16 ?        00:00:00 [ksoftirqd/0]
root          15       2  0 07:16 ?        00:00:00 [rcu_preempt]
root          16       2  0 07:16 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          17       2  0 07:16 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          18       2  0 07:16 ?        00:00:00 [migration/0]
root          19       2  0 07:16 ?        00:00:00 [cpuhp/0]
root          21       2  0 07:16 ?        00:00:00 [kdevtmpfs]
root          22       2  0 07:16 ?        00:00:00 [kworker/R-inet_frag_wq]
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# top -b -n 1 | head -20
top - 07:41:28 up 25 min,  2 users,  load average: 0.01, 0.00, 0.00
Tasks: 110 total,   1 running, 109 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :    957.7 total,    556.0 free,    145.8 used,    255.9 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    679.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0  108228  17716  11196 S   0.0   1.8   0:00.81 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
     11 root      20   0       0      0      0 I   0.0   0.0   0:00.03 kworker/0:1-cgroup_offline
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.04 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:00.04 rcu_preempt
[root@ip-192-168-1-235 ~]# yum install httpd -y
Last metadata expiration check: 1 day, 1:19:35 ago on Wed Oct 28 06:22:10 2026.
Dependencies resolved.
================================================================================================================================
 Package                          Architecture          Version                              Repository                  Size
================================================================================================================================
Installing:
 httpd                            x86_64                2.4.68-1.amzn2023.0.1                amazonlinux                 46 k
Installing dependencies:
 apr                              x86_64                1.7.5-1.amzn2023.0.4                 amazonlinux                129 k
 apr-util                         x86_64                1.6.5-1.amzn2023.0.1                 amazonlinux                100 k
 apr-util-lmdb                    x86_64                1.6.5-1.amzn2023.0.1                 amazonlinux                 15 k
 generic-logos-httpd              noarch                18.0.0-12.amzn2023.0.3               amazonlinux                 19 k
 httpd-core                       x86_64                2.4.68-1.amzn2023.0.1                amazonlinux                1.4 M
 httpd-filesystem                 noarch                2.4.68-1.amzn2023.0.1                amazonlinux                 12 k
 httpd-tools                      x86_64                2.4.68-1.amzn2023.0.1                amazonlinux                 80 k
 libbrotli                        x86_64                1.1.0-1.amzn2023.0.1                 amazonlinux                334 k
 mailcap                          noarch                2.1.49-3.amzn2023.0.3                amazonlinux                 33 k
Installing weak dependencies:
 apr-util-openssl                 x86_64                1.6.5-1.amzn2023.0.1                 amazonlinux                 17 k
 mod_http2                        x86_64                2.0.42-1.amzn2023.0.1                amazonlinux                167 k
 mod_lua                          x86_64                2.4.68-1.amzn2023.0.1                amazonlinux                 59 k

Transaction Summary
================================================================================================================================
Install  13 Packages

Total download size: 2.4 M
Installed size: 7.0 M
Downloading Packages:
(1/13): apr-util-lmdb-1.6.5-1.amzn2023.0.1.x86_64.rpm                            563 kB/s |  15 kB     00:00
(2/13): apr-util-1.6.5-1.amzn2023.0.1.x86_64.rpm                                 3.1 MB/s | 100 kB     00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                                      3.4 MB/s | 129 kB     00:00
(4/13): apr-util-openssl-1.6.5-1.amzn2023.0.1.x86_64.rpm                         625 kB/s |  17 kB     00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm                    708 kB/s |  19 kB     00:00
(6/13): httpd-2.4.68-1.amzn2023.0.1.x86_64.rpm                                   1.8 MB/s |  46 kB     00:00
(7/13): httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch.rpm                        449 kB/s |  12 kB     00:00
(8/13): httpd-core-2.4.68-1.amzn2023.0.1.x86_64.rpm                               33 MB/s | 1.4 MB     00:00
(9/13): httpd-tools-2.4.68-1.amzn2023.0.1.x86_64.rpm                             2.0 MB/s |  80 kB     00:00
(10/13): libbrotli-1.1.0-1.amzn2023.0.1.x86_64.rpm                                13 MB/s | 334 kB     00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm                                1.2 MB/s |  33 kB     00:00
(12/13): mod_http2-2.0.42-1.amzn2023.0.1.x86_64.rpm                              5.1 MB/s | 167 kB     00:00
```

```
(13/13): mod_lua-2.4.68-1.amzn2023.0.1.x86_64.rpm                                                
```

```
   Duration: 53.296s
       Docs: man:httpd.service(8)
    Process: 3996 ExecStart=/usr/sbin/httpd $OPTIONS -DFOREGROUND (code=exited, status=0/SUCCESS)
   Main PID: 3996 (code=exited, status=0/SUCCESS)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
        CPU: 90ms

Oct 29 07:42:03 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Oct 29 07:42:03 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Oct 29 07:42:03 ip-192-168-1-235.ap-south-1.compute.internal httpd[3996]: Server configured, listening on: port 80
Oct 29 07:42:56 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Stopping httpd.service - The Apache HTTP Server...
Oct 29 07:42:57 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: httpd.service: Deactivated successfully.
Oct 29 07:42:57 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Stopped httpd.service - The Apache HTTP Server.
[root@ip-192-168-1-235 ~]# systemctl start httpd
[root@ip-192-168-1-235 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Thu 2026-10-29 07:43:12 UTC; 9s ago
       Docs: man:httpd.service(8)
   Main PID: 4273 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1113)
     Memory: 13.3M
        CPU: 56ms
     CGroup: /system.slice/httpd.service
             ├─4273 /usr/sbin/httpd -DFOREGROUND
             ├─4298 /usr/sbin/httpd -DFOREGROUND
             ├─4301 /usr/sbin/httpd -DFOREGROUND
             ├─4374 /usr/sbin/httpd -DFOREGROUND
             └─4375 /usr/sbin/httpd -DFOREGROUND

Oct 29 07:43:12 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Oct 29 07:43:12 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Oct 29 07:43:12 ip-192-168-1-235.ap-south-1.compute.internal httpd[4273]: Server configured, listening on: port 80
```

Step 13: Automate a daily system report with crontab

Commands:

```
# Create the report script
cat > /usr/local/bin/daily_report.sh << 'EOF'
#!/bin/bash

REPORT_FILE="/var/log/daily_report_$(date +%Y%m%d).txt"

echo "=====================================" > $REPORT_FILE
echo "Daily System Report - $(date)" >> $REPORT_FILE
echo "=====================================" >> $REPORT_FILE
echo "" >> $REPORT_FILE

echo "--- System Info ---" >> $REPORT_FILE
hostname >> $REPORT_FILE
uptime >> $REPORT_FILE
echo "" >> $REPORT_FILE

echo "--- Disk Usage ---" >> $REPORT_FILE
df -h >> $REPORT_FILE
echo "" >> $REPORT_FILE

echo "--- Memory Usage ---" >> $REPORT_FILE
free -h >> $REPORT_FILE
```

```
echo "" >> $REPORT_FILE

echo "--- Top 5 CPU Processes ---" >> $REPORT_FILE
ps aux --sort=-%cpu | head -6 >> $REPORT_FILE
echo "" >> $REPORT_FILE

echo "--- Logged In Users ---" >> $REPORT_FILE
who >> $REPORT_FILE
echo "" >> $REPORT_FILE

echo "--- Recent Error Logs ---" >> $REPORT_FILE
tail -20 /var/log/messages 2>/dev/null || tail -20 /var/log/syslog 2>/dev/null >> $REPORT_FILE

echo "Report saved to: $REPORT_FILE"
EOF

# Make it executable
chmod +x /usr/local/bin/daily_report.sh

# Test run the script manually
/usr/local/bin/daily_report.sh
cat /var/log/daily_report_$(date +%Y%m%d).txt

# Schedule with crontab (run every day at 6 AM)
crontab -e
# Add this line:
# 0 6 * * * /usr/local/bin/daily_report.sh

# Verify crontab
crontab -l
```

Outputs:

```
[root@ip-192-168-1-235 ~]# cat > /usr/local/bin/daily_report.sh << 'EOF'
> #!/bin/bash

REPORT_FILE="/var/log/daily_report_$(date +%Y%m%d).txt"

echo "====================================" > $REPORT_FILE
echo "Daily System Report - $(date)" >> $REPORT_FILE
echo "====================================" >> $REPORT_FILE
echo "" >> $REPORT_FILE

echo "--- System Info ---" >> $REPORT_FILE
hostname >> $REPORT_FILE
uptime >> $REPORT_FILE
echo "" >> $REPORT_FILE

echo "--- Disk Usage ---" >> $REPORT_FILE
df -h >> $REPORT_FILE
echo "" >> $REPORT_FILE

echo "--- Memory Usage ---" >> $REPORT_FILE
free -h >> $REPORT_FILE
echo "" >> $REPORT_FILE
```

```
echo "--- Top 5 CPU Processes ---" >> $REPORT_FILE
ps aux --sort=-%cpu | head -6 >> $REPORT_FILE
echo "" >> $REPORT_FILE

echo "--- Logged In Users ---" >> $REPORT_FILE
who >> $REPORT_FILE
echo "" >> $REPORT_FILE

echo "--- Recent Error Logs ---" >> $REPORT_FILE
tail -20 /var/log/messages 2>/dev/null || tail -20 /var/log/syslog 2>/dev/null >> $REPORT_FILE

echo "Report saved to: $REPORT_FILE"
> EOF
[root@ip-192-168-1-235 ~]# chmod +x /usr/local/bin/daily_report.sh
```

```
[root@ip-192-168-1-235 ~]# /usr/local/bin/daily_report.sh
Report saved to: /var/log/daily_report_20261029.txt
[root@ip-192-168-1-235 ~]# cat /var/log/daily_report_$(date +%Y%m%d).txt
====================================
Daily System Report - Thu Oct 29 07:52:33 UTC 2026
====================================

--- System Info ---
ip-192-168-1-235.ap-south-1.compute.internal
 07:52:33 up 36 min,  2 users,  load average: 0.00, 0.00, 0.00

--- Disk Usage ---
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           479M     0  479M   0% /dev/shm
tmpfs           192M  452K  192M   1% /run
/dev/xvda1      8.0G  1.7G  6.3G  22% /
tmpfs           479M   24K  479M   1% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            96M     0   96M   0% /run/user/1000

--- Memory Usage ---
               total        used        free      shared  buff/cache   available
Mem:           957Mi       181Mi       490Mi       0.0Ki       286Mi       638Mi
Swap:             0B          0B          0B

--- Top 5 CPU Processes ---
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  1.8 108868 18252 ?        Ss   07:16   0:01 /usr/lib/systemd/systemd --switched-root --system --deserialize=32
root           2  0.0  0.0      0     0 ?        S    07:16   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    07:16   0:00 [pool_workqueue_release]
root           4  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-rcu_gp]
root           5  0.0  0.0      0     0 ?        I<   07:16   0:00 [kworker/R-sync_wq]

--- Logged In Users ---
ec2-user pts/0        2026-10-29 07:48 (152.58.46.241)
ec2-user pts/1        2026-10-29 07:48 (152.58.46.241)
```

```
--- Recent Error Logs ---
[root@ip-192-168-1-235 ~]# crontab -e
-bash: crontab: command not found
[root@ip-192-168-1-235 ~]# dnf install -y cronie
Last metadata expiration check: 1 day, 1:32:26 ago on Wed Oct 28 06:22:10 2026.
Dependencies resolved.
================================================================================================================================================
 Package                        Architecture           Version                                  Repository                     Size
================================================================================================================================================
Installing:
 cronie                         x86_64                 1.5.7-1.amzn2023.0.2                     amazonlinux                   115 k
Installing dependencies:
 cronie-anacron                 x86_64                 1.5.7-1.amzn2023.0.2                     amazonlinux                    32 k

Transaction Summary
================================================================================================================================================
Install  2 Packages

Total download size: 147 k
Installed size: 341 k
Downloading Packages:
(1/2): cronie-1.5.7-1.amzn2023.0.2.x86_64.rpm                                   4.5 MB/s | 115 kB     00:00
(2/2): cronie-anacron-1.5.7-1.amzn2023.0.2.x86_64.rpm                           1.2 MB/s |  32 kB     00:00
------------------------------------------------------------------------------------------------------------------------------------------------
Total                                                                           1.8 MB/s | 147 kB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                                       1/1
  Installing       : cronie-anacron-1.5.7-1.amzn2023.0.2.x86_64                                                                            1/2
  Running scriptlet: cronie-anacron-1.5.7-1.amzn2023.0.2.x86_64                                                                            1/2
  Installing       : cronie-1.5.7-1.amzn2023.0.2.x86_64                                                                                    2/2
  Running scriptlet: cronie-1.5.7-1.amzn2023.0.2.x86_64                                                                                    2/2
Created symlink /etc/systemd/system/multi-user.target.wants/crond.service → /usr/lib/systemd/system/crond.service.
```

```
    Release notes:
     https://docs.aws.amazon.com/linux/al2023/release-notes/relnotes-2023.12.20261028.html

================================================================================================================================

Installed:
  cronie-1.5.7-1.amzn2023.0.2.x86_64                          cronie-anacron-1.5.7-1.amzn2023.0.2.x86_64

Complete!
[root@ip-192-168-1-235 ~]# systemctl status crond
○ crond.service - Command Scheduler
     Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: enabled)
     Active: inactive (dead)
[root@ip-192-168-1-235 ~]# systemctl enable --now crond
[root@ip-192-168-1-235 ~]# systemctl status crond
● crond.service - Command Scheduler
     Loaded: loaded (/usr/lib/systemd/system/crond.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-10-29 07:54:57 UTC; 2s ago
   Main PID: 5421 (crond)
      Tasks: 1 (limit: 1113)
     Memory: 1.1M
        CPU: 5ms
     CGroup: /system.slice/crond.service
             └─5421 /usr/sbin/crond -n

Oct 29 07:54:57 ip-192-168-1-235.ap-south-1.compute.internal crond[5421]: (CRON) STARTUP (1.5.7)
Oct 29 07:54:57 ip-192-168-1-235.ap-south-1.compute.internal systemd[1]: Started crond.service - Command Scheduler.
Oct 29 07:54:57 ip-192-168-1-235.ap-south-1.compute.internal crond[5421]: (CRON) INFO (Syslog will be used instead of sendmail.)
Oct 29 07:54:57 ip-192-168-1-235.ap-south-1.compute.internal crond[5421]: (CRON) INFO (RANDOM_DELAY will be scaled with factor 31% i>
Oct 29 07:54:57 ip-192-168-1-235.ap-south-1.compute.internal crond[5421]: (CRON) INFO (running with inotify support)

[root@ip-192-168-1-235 ~]# crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
[root@ip-192-168-1-235 ~]# crontab -l
# 0 6 * * * /usr/local/bin/daily_report.sh
```

Step 14: Environment variables

Commands:

View current environment variables
env
printenv PATH
printenv HOME

Create custom environment variables
export PROJECT_DIR="/project"
export ADMIN_EMAIL="admin@company.com"
export SERVER_ENV="production"

Verify
echo $PROJECT_DIR
echo $ADMIN_EMAIL
echo $SERVER_ENV

Make permanent (add to profile)
echo 'export PROJECT_DIR="/project"' >> ~/.bashrc
echo 'export ADMIN_EMAIL="admin@company.com"' >> ~/.bashrc
source ~/.bashrc

Verify after reload

echo $PROJECT_DIR

Outputs:

```
[root@ip-192-168-1-235 ~]# env
SHELL=/bin/bash
HISTCONTROL=ignoredups
SYSTEMD_COLORS=false
HISTSIZE=1000
HOSTNAME=ip-192-168-1-235.ap-south-1.compute.internal
PWD=/root
LOGNAME=root
HOME=/root
LANG=C.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=01;37;41:su=37;41:sg=30;43:ca=30;41:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=01;36:*.au=01;36:*.flac=01;36:*.m4a=01;36:*.mid=01;36:*.midi=01;36:*.mka=01;36:*.mp3=01;36:*.mpc=01;36:*.ogg=01;36:*.ra=01;36:*.wav=01;36:*.oga=01;36:*.opus=01;36:*.spx=01;36:*.xspf=01;36:
TERM=xterm-256color
LESSOPEN=||/usr/bin/lesspipe.sh %s
USER=root
SHLVL=1
S_COLORS=auto
which_declare=declare -f
PATH=/root/.local/bin:/root/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
MAIL=/var/spool/mail/root
BASH_FUNC_which%%=() {  ( alias;
 eval ${which_declare} ) | /usr/bin/which --tty-only --read-alias --read-functions --show-tilde --show-dot "$@"
}
_=/usr/bin/env
[root@ip-192-168-1-235 ~]# printenv PATH
/root/.local/bin:/root/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin
[root@ip-192-168-1-235 ~]# printenv HOME
```

```
[root@ip-192-168-1-235 ~]# printenv HOME
/root
[root@ip-192-168-1-235 ~]# export PROJECT_DIR="/project"
[root@ip-192-168-1-235 ~]# export ADMIN_EMAIL="admin@company.com"
[root@ip-192-168-1-235 ~]# export SERVER_ENV="production"
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# echo $PROJECT_DIR
/project
[root@ip-192-168-1-235 ~]# echo $ADMIN_EMAIL
admin@company.com
[root@ip-192-168-1-235 ~]# echo $SERVER_ENV
production
[root@ip-192-168-1-235 ~]# echo 'export PROJECT_DIR="/project"' >> ~/.bashrc
[root@ip-192-168-1-235 ~]# echo 'export ADMIN_EMAIL="admin@company.com"' >> ~/.bashrc
[root@ip-192-168-1-235 ~]# source ~/.bashrc
[root@ip-192-168-1-235 ~]# echo $PROJECT_DIR
/project
[root@ip-192-168-1-235 ~]#
```

Step 15: Final system verification

Commands:

Show all users created
cat /etc/passwd | grep -E "alice|bob|charlie"

Show all groups
cat /etc/group | grep -E "developers|managers"

Show directory permissions
ls -la /project/

Show crontab
crontab -l

Show running services
systemctl list-units --type=service --state=running

Show disk usage
df -h

Show memory
free -h

Show system uptime
uptime

Show who is logged in
who
w
last | head -10

Outputs:

```
[root@ip-192-168-1-235 ~]# cat /etc/passwd | grep -E "alice|bob|charlie"
alice:x:1001:1001::/home/alice:/bin/bash
bob:x:1002:1001::/home/bob:/bin/bash
charlie:x:1003:1002::/home/charlie:/bin/bash
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# cat /etc/group | grep -E "developers|managers"
developers:x:1001:
managers:x:1002:
[root@ip-192-168-1-235 ~]# ls -la /project/
total 0
drwxr-xr-x.  5 root root        46 Oct 28 07:17 .
dr-xr-xr-x. 19 root root       252 Oct 28 07:17 ..
drwxrwx---+  2 root developers  28 Oct 28 07:48 dev
drwxrwx---.  2 root managers    32 Oct 28 07:50 manager
drwxrwxrwx.  2 root root       102 Oct 28 11:20 shared
[root@ip-192-168-1-235 ~]#
[root@ip-192-168-1-235 ~]# crontab -l
# 0 6 * * * /usr/local/bin/daily_report.sh


[root@ip-192-168-1-235 ~]# systemctl list-units --type=service --state=running
  UNIT                        LOAD   ACTIVE SUB     DESCRIPTION
  acpid.service               loaded active running ACPI Event Daemon
  amazon-ssm-agent.service    loaded active running amazon-ssm-agent
  atd.service                 loaded active running Deferred execution scheduler
  auditd.service              loaded active running Security Auditing Service
  chronyd.service             loaded active running NTP client/server
  crond.service               loaded active running Command Scheduler
  dbus-broker.service         loaded active running D-Bus System Message Bus
  getty@tty1.service          loaded active running Getty on tty1
  gssproxy.service            loaded active running GSSAPI Proxy Daemon
  httpd.service               loaded active running The Apache HTTP Server
  libstoragemgmt.service      loaded active running libstoragemgmt plug-in server daemon
  serial-getty@ttyS0.service loaded active running Serial Getty on ttyS0
  sshd.service                loaded active running OpenSSH server daemon
  systemd-homed.service       loaded active running Home Area Manager
  systemd-journald.service    loaded active running Journal Service
  systemd-logind.service      loaded active running User Login Management
  systemd-networkd.service    loaded active running Network Configuration
  systemd-resolved.service    loaded active running Network Name Resolution
  systemd-udevd.service       loaded active running Rule-based Manager for Device Events and Files
  systemd-userdbd.service     loaded active running User Database Manager
  user@1000.service           loaded active running User Manager for UID 1000
```

```
LOAD   = Reflects whether the unit definition was properly loaded.
ACTIVE = The high-level unit activation state, i.e. generalization of SUB.
SUB    = The low-level unit activation state, values depend on unit type.
21 loaded units listed.
[root@ip-192-168-1-235 ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M     0  4.0M   0% /dev
tmpfs           479M     0  479M   0% /dev/shm
tmpfs           192M  460K  192M   1% /run
/dev/xvda1      8.0G  1.7G  6.3G  22% /
tmpfs           479M   24K  479M   1% /tmp
/dev/xvda128     10M  1.3M  8.7M  13% /boot/efi
tmpfs            96M     0   96M   0% /run/user/1000
[root@ip-192-168-1-235 ~]# free -h
               total        used        free      shared  buff/cache   available
Mem:           957Mi       184Mi       480Mi       0.0Ki       292Mi       634Mi
Swap:             0B          0B          0B
[root@ip-192-168-1-235 ~]# uptime
 08:07:56 up 51 min,  2 users,  load average: 0.00, 0.00, 0.00
[root@ip-192-168-1-235 ~]# who
ec2-user pts/0        2026-10-29 08:03 (152.58.46.241)
ec2-user pts/1        2026-10-29 08:03 (152.58.46.241)
[root@ip-192-168-1-235 ~]# w
 08:08:08 up 51 min,  2 users,  load average: 0.00, 0.00, 0.00
USER     TTY        LOGIN@   IDLE   JCPU   PCPU WHAT
ec2-user pts/0     08:03    0.00s  0.02s  0.01s sshd-session: ec2-user [priv]
ec2-user pts/1     08:03    0.00s  0.02s  0.01s sudo su -
[root@ip-192-168-1-235 ~]# last | head -10
ec2-user pts/0        152.58.46.241    Thu Oct 29 08:03   still logged in
ec2-user pts/0        152.58.46.241    Thu Oct 29 07:48 - 08:01  (00:12)
ec2-user pts/0        152.58.46.241    Thu Oct 29 07:39 - 07:48  (00:08)
ec2-user pts/0        152.58.46.241    Thu Oct 29 07:29 - 07:38  (00:09)
ec2-user pts/0        152.58.46.241    Thu Oct 29 07:19 - 07:29  (00:09)
reboot   system boot  6.18.51-120.163. Thu Oct 29 07:16   still running
ec2-user pts/0        152.58.43.224    Wed Oct 28 11:09 - 11:33  (00:24)
reboot   system boot  6.18.51-120.163. Wed Oct 28 11:06 - 15:14  (04:07)
```

```
reboot   system boot  6.18.51-120.163. Thu Oct 29 07:16   still running
ec2-user pts/0        152.58.43.224    Wed Oct 28 11:09 - 11:33  (00:24)
reboot   system boot  6.18.51-120.163. Wed Oct 28 11:06 - 15:14  (04:07)
ec2-user pts/0        152.58.43.211    Wed Oct 28 07:47 - 07:54  (00:07)
ec2-user pts/0        152.58.43.211    Wed Oct 28 07:09 - 07:42  (00:33)
[root@ip-192-168-1-235 ~]#
```

THANK YOU!!